

Lecture 1 Notes:

What Is So Interesting About Mathematics for a Philosopher?

Philosophy of Mathematics

Central Theme

Mathematics is philosophically interesting because it appears to be abstract, necessary, and knowable by reason alone, while also being indispensable to empirical science and ordinary inquiry.

Mathematics = abstract reason + scientific applicability

The guiding question for the lecture:

Who has authority over mathematics?

Philosophers? Mathematicians? Science? Mathematical practice itself?

1. Mathematics as a Philosophical Test Case

Mathematics has attracted philosophers throughout history. Plato treated mathematics as training for grasping reality as it is, rather than merely as it appears. Many major figures in philosophy were also deeply engaged with mathematics or logic: Descartes, Leibniz, Frege, Russell, Hilbert, Gödel, Tarski, and others.

Mathematics is a test case for several major philosophical problems:

- Knowledge : How do we know mathematical truths?
- Reality : What are numbers, sets, functions, spaces?
- Language : What do mathematical terms refer to?
- Logic : What counts as valid reasoning?
- Science : Why is mathematics so useful in empirical inquiry?

Teaching Point

Philosophy of mathematics is not just “philosophers talking about numbers.” It is a way into central questions about knowledge, reality, language, logic, and science.

Whiteboard

Mathematics $\longrightarrow$ $\left\{ \begin{array}{l} \text{Knowledge} \\ \text{Reality} \\ \text{Language} \\ \text{Logic} \\ \text{Science} \end{array} \right.$

2. Rationalism and Empiricism

Mathematics seems to provide knowledge that is certain, necessary, and independent of sense experience. This makes it central to the dispute between rationalism and empiricism.

Rationalism	Empiricism
Reason is a source of knowledge	Experience is the source of knowledge
Mathematics is a model	Mathematics is a problem
Plato, Descartes, Leibniz	Aristotle, Locke, Hume, Mill

For the rationalist, mathematics shows what genuine knowledge looks like: proof, necessity, certainty. For the empiricist, mathematics is a challenge: if knowledge comes from experience, how do we explain mathematical knowledge?

Teaching Point

Mathematics appears to be a paradigm case of *a priori* knowledge: knowledge prior to, or independent of, experience.

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Mathematics = paradigm of *a priori* knowledge?

$2 + 3 = 5$ known by proof/reason, not experiment?

3. Mathematics and the Sciences

Mathematics is not merely abstract. It is also indispensable to the natural and social sciences. Physics, engineering, economics, computer science, statistics, and many parts of biology depend on mathematical representation.

This creates a central puzzle:

Why does abstract mathematics apply to the physical world?

Galileo's metaphor is useful here: the book of nature is written in the language of mathematics.

Teaching Point

Mathematics seems independent of experience, yet it is one of our most powerful tools for understanding experience. This tension is one of the deepest motivations for philosophy of mathematics.

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Abstract mathematics $\longrightarrow$ empirical science

Puzzle: Why does mathematics apply?

4. Logic as a Bridge Between Mathematics and Philosophy

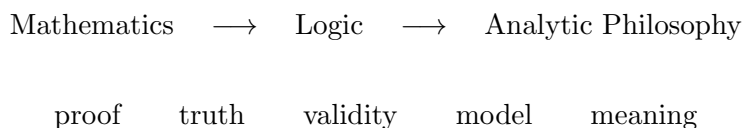
Modern formal logic developed largely through reflection on mathematical reasoning. Figures such as Boole, Frege, Russell, Hilbert, Gödel, Tarski, and Church were interested in the foundations of mathematics. But the tools they developed became central to analytic philosophy more generally.

Logic connects mathematics to questions about proof, truth, consequence, validity, meaning, and reference.

Teaching Point

Mathematics did not merely give philosophers something to study. It helped shape the formal tools philosophers use to study many other things.

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5. Philosophy and Mathematics: Chicken or Egg?

The central methodological question of the chapter is whether philosophy should govern mathematics, or whether mathematics should govern philosophy.

Does philosophy determine mathematics, or does mathematics determine philosophy?

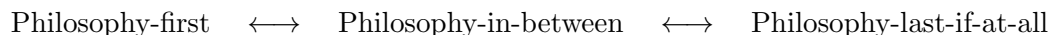
The chapter presents three broad positions:

Philosophy-first	Middle view	Philosophy-last
Philosophy governs	Philosophy interprets	Mathematics governs
First principles	Mutual illumination	Practice first
Revisionary	No domination	Anti-revisionist

Teaching Point

The chapter is not only about mathematics. It is also about disciplinary authority. Who gets to say what counts as legitimate mathematics?

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6. Plato: Philosophy Pressuring Geometry

Plato held that the true objects of mathematics are eternal, unchanging, and ideal. Geometry is not really about the imperfect figures we draw. It is about ideal objects grasped by reason.

But geometers often use constructive language:

draw a line move a figure construct a triangle

Plato objects that this language is misleading. If mathematical objects are eternal and unchanging, then they are not literally drawn, moved, or constructed.

Teaching Point

Plato is an example of philosophy-first. His metaphysics of mathematical objects leads him to criticize ordinary mathematical language.

Whiteboard

Plato

Mathematical objects are eternal and unchanging.

$\therefore$ “draw a line” is philosophically misleading.

7. Intuitionism: Philosophy Pressuring Logic

Intuitionists such as Brouwer and Heyting hold that mathematics concerns mental construction. To say that a mathematical object exists is to say, roughly, that it can be constructed.

This puts pressure on classical logic, especially the law of excluded middle:

$$P \vee \neg P$$

Classically, the following inference is valid:

$$\neg \forall x P(x) \quad \Rightarrow \quad \exists x \neg P(x)$$

But the intuitionist rejects this as a general principle. To assert an existential claim, one must be able to construct an example.

Teaching Point

For intuitionists, logic depends on the meaning of mathematical existence. If existence means constructibility, then some classical principles become suspect.

Whiteboard

Intuitionism

Mathematics = mental construction

To exist = to be constructible

$$\neg \forall x P(x) \quad \not\Rightarrow_{\text{intuitionist}} \quad \exists x \neg P(x)$$

8. Impredicative Definitions: Poincaré and Gödel

A definition is impredicative when it defines an object by reference to a totality that already contains that object.

Impredicative definition = definition using a totality containing the thing defined

Poincaré objected that such definitions are circular. If mathematical objects are constructed, then one cannot construct an object by appealing to a totality that already contains it.

Gödel defended impredicative definitions from a realist perspective. If mathematical objects exist independently of us, then definitions do not create them. They describe or identify objects that already exist.

Teaching Point

Whether an impredicative definition is legitimate depends partly on whether definitions are understood as recipes for constructing objects or descriptions of already existing objects.

Whiteboard

Poincaré	Gödel
Objects constructed	Objects discovered
Definitions as recipes	Definitions as descriptions
Impredicativity circular	Impredicativity legitimate

9. Anti-Revisionism: Mathematical Practice Pushes Back

The chapter argues that philosophy-first does not fit the history of mathematics very well.

Classical logic and impredicative definitions are deeply entrenched in modern mathematics. But mathematicians did not generally accept them because they first adopted a realist metaphysics. Rather, these methods became accepted because they were mathematically fruitful.

Classical logic + impredicative definitions = central to modern mathematics

Teaching Point

Anti-revisionism says that successful mathematical practice has default authority. A philosophical argument must be very strong before it can overturn established mathematics.

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Anti-revisionism

Successful mathematical practice has default authority.

If philosophy conflicts with successful mathematics,

maybe philosophy is the problem.

10. Dummett: Language and Mathematical Revision

Michael Dummett gives another example of a philosophical argument with revisionary consequences. He argues from considerations about language, meaning, learnability, and communication.

If meaning must be connected to use and verification, then classical mathematics may be problematic when it appeals to truths that outrun our capacity to recognize them.

This pushes Dummett toward intuitionistic logic.

Teaching Point

Dummett raises the question of whether a theory of meaning can constrain mathematical logic.

Whiteboard

Dummett

Meaning $\longrightarrow$ use, learnability, communication

$\therefore$ classical logic is not generally valid?

Revise mathematics or reject the philosophy of language?

11. Naturalism: No First Philosophy

The final section introduces naturalism. Naturalism rejects the idea that philosophy stands outside science and mathematics as a superior tribunal.

For Quine, philosophy begins within our inherited scientific worldview. We do not justify science or mathematics from an external standpoint. We revise our beliefs from within our overall web of belief.

The image is Neurath's boat: we rebuild the ship while already at sea.

Teaching Point

Naturalism rejects philosophy-first. Philosophy is not prior to inquiry. It is continuous with inquiry.

Whiteboard

Naturalism

No first philosophy

Philosophy is continuous with inquiry

Neurath's boat: repair the ship while at sea

12. Quine and Maddy

Quine and Maddy both reject philosophy-first, but they differ over the status of mathematics.

For Quine, mathematics is accepted because of its role in empirical science. Mathematics is part of the scientific web of belief.

For Maddy, mathematics has its own methods, standards, and internal goals. Mathematics should not need external justification from either philosophy or immediate scientific application.

Quine	Maddy
Science-centered naturalism	Mathematics-centered naturalism
Mathematics justified by science	Mathematics has its own standards
Web of belief	Seams in the web
Empirical role matters	Mathematical practice matters

Teaching Point

Quine rejects philosophy-first, but may still pressure mathematics from the standpoint of science. Maddy gives more autonomy to mathematics itself.

Whiteboard

Quine: mathematics earns its place through science.

Maddy: mathematics has autonomous standards.

Conclusion: Why Mathematics Matters Philosophically

Mathematics is philosophically important because it combines features that are difficult to hold together.

Feature of Mathematics	Philosophical Problem
Abstract	What is mathematics about?
Necessary	How can we know necessary truths?
A priori	How does this fit with empiricism?
Logical	What principles of reasoning are legitimate?
Applicable	Why does mathematics describe the world?
Autonomous	Can philosophy criticize mathematical practice?

Final Framing

To teach philosophy of mathematics well, we should present mathematics not merely as a body of theorems, but as a philosophical pressure point.

Mathematics appears to be the purest exercise of reason, yet it is indispensable to our empirical knowledge of the world. It seems autonomous, yet it raises questions that reach into epistemology, metaphysics, logic, language, and philosophy of science.

Philosophy of mathematics interprets mathematics and its place in inquiry.

Discussion Questions

1. Why has mathematics been so attractive to rationalist philosophers?
2. Why is mathematics difficult for empiricism?
3. Should philosophers ever tell mathematicians to change their methods?
4. Does mathematical existence require construction?
5. Are mathematical definitions recipes or descriptions?
6. Is mathematics justified by its role in science?
7. Does pure mathematics need external justification?
8. Is philosophy of mathematics supposed to ground mathematics, criticize it, or interpret it?

Key Terms

Rationalism The view that reason is a fundamental source of knowledge.

Empiricism The view that sense experience is the primary source of knowledge.

A priori knowledge Knowledge independent of sense experience.

Philosophy-first The view that philosophy supplies first principles that govern mathematical practice.

Philosophy-last-if-at-all The view that mathematics is autonomous and philosophy merely describes it afterward, if it has any role at all.

Anti-revisionism The view that successful mathematical practice should not be revised merely because of external philosophical arguments.

Intuitionism The view that mathematics concerns mental construction and that existence requires constructibility.

Law of excluded middle The classical principle that for any proposition P , either P or $\neg P$ holds.

Impredicative definition A definition that characterizes an object by reference to a totality containing that object.

Realism The view that mathematical objects exist independently of minds, language, or constructions.

Naturalism The view that philosophy is continuous with successful inquiry rather than an external foundation for it.